STRENGTHENING U.S. AQUACULTURE THROUGH BUSINESS SUPPORT

Seven projects based in Florida, Kansas, North Carolina, Puerto Rico, Virginia, and Washington were selected for a total of $4,235,440 in fiscal year 2025 funding, towards strengthening U.S. aquaculture. The overall goal of the projects is to support new aquaculture businesses and enhance the output, efficiency, and profitability of U.S. aquaculture businesses. Learn more about these projects below.

FLORIDA

The development of a cost-effective and scalable protected species monitoring plan for the Velella Epsilon and future finfish aquaculture projects in the Gulf of America

Lead: New College Of Florida
Principal Investigators: Athena Rycyk (New College of Florida), Jessica Powell (NOAA Fisheries), Neil Sims (Ocean Era, Inc.), Lindsey Feldman (NOAA Fisheries Southeast and Alaska Regional Office), Dennis Peters (Gulfstream Aquaculture, LLC)
Grant Amount: $331,274
This project will develop a cost-effective and scalable protected species monitoring plan to assess potential effects to Endangered Species Act-listed species and marine mammals from offshore aquaculture operations. Passive acoustic monitoring devices will be deployed and visual surveys will be performed at an offshore fish farming site, both before and after deployment of a net pen. This project will not only assess potential impacts to protected species but also review which monitoring methods are most cost-effective for aquaculture business operators while still allowing for an accurate assessment of potential effects.

KANSAS

Deterring birds to enhance oyster farm sustainability: Evaluating deterrents, equipment design, and economic impacts

Lead: Kansas State University

Principal Investigators: Joe Gerken (Kansas State University), Jesse Tack (Kansas State University), Heather Ketcham (Ketcham Supply Co., Inc.)

Grant Amount: $662,218

This project will address the presence of birds at oyster farms, particularly farms that use floating culture gear, as feces from birds can decrease product quality and result in human safety concerns due to harmful bacteria. Work will identify techniques to deter birds, examine the economic cost and benefits of deterrence methods, and create materials to help producers utilize research findings. Findings of this study will help regulators create policy and best practices to maintain human health.

NORTH CAROLINA

Addressing emerging needs of North Carolina's shellfish aquaculture industry to support its continued growth

Lead: North Carolina Coastal Federation

Principal Investigators: Ana Zivanovic-Nenadovic (North Carolina Coastal Federation, Inc.), Jane Harrison (North Carolina Sea Grant)

Grant Amount: $529,140

This project will address the emerging needs of North Carolina's rapidly growing shellfish aquaculture industry by updating and expanding key elements of the industry's existing strategic plan, investing in essential infrastructure, and enhancing outreach to improve public understanding of the industry's importance. Work will establish improved guidelines for sustainable shellfish farm siting; support a stable, reliable supply of North Carolina-specific oyster seed; and develop a sustainable operational model and blueprint for replication in other locations. Results will be used to improve public understanding of shellfish aquaculture's economic and environmental benefits through outreach and tourism initiatives.

PUERTO RICO

Bringing a commercial open ocean aquaculture farm to Puerto Rico

Lead: Innovasea Systems, Inc.
Principal Investigator: Tyler Sclodnick (InnovaSea Systems, Inc.)
Grant Amount: $395,713
This project will advance the planning and permitting of an open ocean net pen farm off the west coast of Puerto Rico to reduce the risks associated with starting a new farm. Work will consist of conducting ocean surveys, deploying environmental sensors and organizing meetings with key partners and potential suppliers on the island. This information will be utilized to perform a site selection and project feasibility analysis. In addition, a socio-economic analysis and partner engagement efforts will be conducted, which will foster community collaboration and build social license for sustainable aquaculture projects in Puerto Rico.

VIRGINIA

Go big or go home? Detailing the drivers and outcomes of U.S. shellfish aquaculture industry attrition and growth

Lead: Virginia Institute of Marine Science
Principal Investigators: Adriane Michaelis (Virginia Institute of Marine Science), Eric Herbst (North Carolina Sea Grant), Karen Hudson (Virginia Institute of Marine Science), Josh Reitsma (Cape Cod Cooperative Extension and Woods Hole Oceanographic Institution Sea Grant), Andrew Scheld (Virginia Institute of Marine Science), Leslie Sturmer (University of Florida/IFAS Extension), Bill Walton (Virginia Institute of Marine Science), Shelby White (Virginia Institute of Marine Science)
Grant Amount: $593,074
The goal of this project is to better understand United States shellfish aquaculture industry growth patterns, with a focus on business-level decision-making related to entering, exiting, and expanding within the industry. Work will be conducted to investigate likely drivers of industry change as well as the likelihood of business success given different scales of operation and state-specific characteristics, such as restrictions on total farm acreage. This information will better inform current and prospective shellfish growers about the effect of scale on their economic sustainability, allowing them to position themselves for long-term success.

WASHINGTON

Sea of Opportunity – Strengthening the shellfish industry workforce in Washington

Lead: Washington Sea Grant
Principal Investigators: Melissa Poe (Washington Sea Grant, mpoe@uw.edu), Nicole Naar (Washington Sea Grant)
Grant Amount: $744,451

This project will provide workforce development to the shellfish aquaculture industry in Washington state. The industry is part of a safe and sustainable seafood economy that supports meaningful family wage jobs in rural communities. The team will optimize a pilot workforce development program called Tide's Out. Expanding Tide's Out to serve a greater range of shellfish industry businesses statewide and engage middle and high school students will improve workforce readiness. The team will also create communication materials that improve social license for shellfish aquaculture, enhance perceptions of aquaculture careers, and evaluate program effectiveness.

WASHINGTON

Valuing shellfish aquaculture as estuarine habitat for regulatory compliance, social license and consumer education

Lead: Pacific Shellfish Institute
Principal Investigators: Bobbi Hudson (Pacific Shellfish Institute), Katie Houle (Pacific Shellfish Institute), Fiona Boardman (Pacific Shellfish Institute), Brett Dumbauld (USDA Agricultural Research Service), James Losee (Wild Salmon Center)
Grant Amount: $979,570
This project will document and communicate the ecosystem services provided by shellfish farms—particularly their role as estuarine habitat for fish, crab, birds, and marine mammals. In partnership with shellfish farms, the team will use video data, eDNA, acoustic telemetry, and observations to study the biodiversity and behavior of native species occurring at Pacific Coast shellfish farms across seasons, culture methods, and growing regions. Communication tools and strategies to engage with the aquaculture community and the general public will also be created. The team will also develop ecosystem services and aquaculture career lessons for middle and high school students. Collectively, the proposed research and outreach will support new and existing shellfish businesses regarding regulatory compliance, social license, and consumer education.

ENHANCING PRODUCTION OF AQUACULTURE SPECIES

Three projects based in Hawai‘i, Texas, and Virginia were selected for a total of $2,583,009 in fiscal year 2025 funding, towards developing and refining methods, protocols, techniques, and/or strategies to enhance the production of one or more life stages of aquaculture species. The overall goal of these projects is to improve the efficiency, output, and profitability of U.S. aquaculture businesses. Learn more about these projects below.

HAWAIʻI

Resolving the environmental impediments of sea cucumber production in Hawaiʻi through the integration of spawning enhancement and restorative aquaculture strategies

Lead: University Of Hawaiʻi At Mānoa/Hawaiʻi Sea Grant
Principal Investigators: Andre Seale (University of Hawaiʻi at Mānoa), Bradley ʻKaiʼ Fox (Hawaiʻi Sea Grant), Herb Lee (Pacific American Foundation), Darren K. Okimoto (Hawaiʻi Sea Grant)
Grant Amount: $1,136,041
The goal of this project is to improve the spawning capacity of sea cucumbers as a means to enhance the aquaculture of this species for human consumption and restoration efforts. One of the largest constraints to increasing the production of sea cucumbers is their lack of reliable spawning under different environmental conditions. Results from this work will provide aquaculture practitioners with knowledge to induce spawning and control propagation of native Hawaiian sea cucumbers, thereby supporting increased food production and the restoration of wild populations. This will also diminish the reliance on harvesting wild sea cucumber stocks, which is a major threat facing sea cucumbers worldwide.

TEXAS

Improving the genomic toolbox for polyploid oysters

Lead: Texas A&M University-Corpus Christi
Principal Investigators: Christopher Hollenbeck (Texas A&M University - Corpus Christi), David Portnoy (Texas A&M University - Corpus Christi), Scott Rikard (Auburn University), Andrea Tarnecki (Auburn University), Jessica Small (Virginia Institute of Marine Science)
Grant Amount: $475,795
The goal of this project is to improve the selective breeding of polyploid eastern oysters, since a considerable proportion of eastern oyster aquaculture in the U.S. consists of the grow-out of triploid individuals produced using tetraploid broodstock. Work will design and validate polyploid genomic tools and genetic analysis protocols, and use genomic tools to investigate the issue of unexplained triploid mortality, a pressing concern for the industry. By developing technology for genetic improvement of eastern oysters, the project will increase annual production and profitability of the U.S. oyster aquaculture industry.

VIRGINIA

Innovative avian deterrence strategies for sustainable aquaculture

Lead: Old Dominion University
Principal Investigators: John Shull (Old Dominion University), John Swaddle (College of William and Mary), Bill Walton (Virginia Institute of Marine Science), Hector Garcia (Old Dominion University), Yiannis Papelis (Old Dominion University)
Grant Amount: $971,173
This project will address bird activity at oyster farms, particularly those using floating culture gear, and the resulting impacts of bird feces on oyster health, product integrity, and damage to equipment. A dynamic, AI-powered 'Sonic Net' will be developed that integrates real-time video monitoring, machine learning bird detection, and an adaptive acoustic deterrence model. The system will detect bird presence and adjust acoustic output based on likely species and environmental conditions. This adaptive learning approach is designed to reduce habituation and improve deterrence effectiveness over time. The end result is intended to be smart, cost-effective, and a commercially viable system that supports regulatory compliance while providing flexible, farm-specific deterrence solutions.

AQUACULTURE INTERNSHIPS PROGRAM

Four projects based in California, Hawai'i, Louisiana, and New Jersey were selected for a total of $1,493,911 in fiscal year 2025 funding, towards enhancing workforce development of the U.S. aquaculture industry through internships. The internships will provide commercial aquaculture experience for students attending high school, or undergraduate or graduate school, including community colleges, and other interested individuals at partnering aquaculture operations. Learn more about these projects below.

CALIFORNIA SEA GRANT

A community college aquaculture internship program to help ready Southern California for a growing aquaculture industry

Principal Investigators: Theresa Talley (California Sea Grant), Simona Augyte (California Sea Grant), Thomas Tubon (MiraCosta College)
Grant Amount: $293,073
This project will launch a paid, 100-hour aquaculture internship program that provides on-the-job training at partner facilities for students enrolled in an ongoing community college aquaculture training program. The program will result in well-trained, new aquaculture staff who are connected to the aquaculture industry and related operations, including ports, public aquariums, and systems supply and design businesses, across the region. This model program and its training materials will be able to be further adapted and expanded to other regions in California and beyond.

HAWAIʻI SEA GRANT

Hawaiʻi-Pacific Regional Aquaculture Internship Program 2025-2027

Principal Investigators: Anne Rosa (Hawaiʻi Sea Grant), Cherie Kauahi (Hawaiʻi Sea Grant), Darren K. Okimoto (Hawaiʻi Sea Grant), Bradley ʻKaiʼ Fox (Hawaiʻi Sea Grant)
Grant Amount: $476,135
This internship program will provide direct work experience in the aquaculture industry and academic institutions for students and community members across Hawaiʻi and the U.S.-affiliated Pacific Island region. To provide these experiences, the project will include partnerships with businesses, non-profits, and academic institutions for high school and undergraduate students, as well as early-career adults. The project will increase workforce capacity in the aquaculture sector, broaden community participation in the industry, and strengthen collaborations between educational institutions, non-profits, and businesses.

LOUISIANA SEA GRANT

Louisiana Sea Grant aquaculture internship pilot program

Principal Investigators: Julie Lively (Louisiana Sea Grant), Wood Oglesby (Louisiana Sea Grant), Haley Gambill (Louisiana Sea Grant), Elizabeth Robinson (Louisiana Sea Grant),
Danielle Dilullo (Louisiana Sea Grant)
Grant Amount: $434,000
This project aims to address Louisiana's aging aquaculture workforce by creating internships across multiple aquaculture sectors throughout the state. The established internships will be 12 or 24 weeks in duration and offered in multiple aquaculture sectors, including alligator, crawfish, aquaponics, oyster, catfish, and other fish species. The program will be focused on students from high schools, community colleges, and four-year institutions, as well as members of the commercial fishing community and the general public. This internship program will increase the workforce capacity of the varied Louisiana aquaculture sector, thereby contributing to industry productivity and profitability.

NEW JERSEY SEA GRANT CONSORTIUM

Strengthening the New Jersey Apprenticeship in Shellfish Aquaculture Program (ASAP)

Principal Investigators: Peter Rowe (New Jersey Sea Grant Consortium, prowe@njseagrant.org), Diana Burich (New Jersey Sea Grant Consortium)
Grant Amount: $290,703
This project will continue the established Apprenticeship in Shellfish Aquaculture Program (ASAP), which provides high school students with the knowledge, skills, and inspiration to pursue aquaculture-related career paths. The program will develop an aquaculture literacy curriculum with local school districts, recruit apprentices, train students in core skills at an aquaculture "bootcamp", connect students with paid summer apprenticeships at shellfish farms, support apprentices with professional development and networking opportunities, and provide additional aquaculture job training for vocational school students. The program will result in a more resilient and productive aquaculture industry and provide communities with increased knowledge of and appreciation for aquaculture careers.

AQUACULTURE LEGAL, REGULATORY, AND POLICY GRANTS

Three projects based in California, Mississippi, and Virginia were selected for a total of $554,830 in fiscal year 2025 funding, towards addressing legal, regulatory, and policy topics relevant to the U.S. aquaculture community. Learn more about these projects below.

CALIFORNIA SEA GRANT

Facilitation for aquaculture interagency and industry coordination

Principal Investigator: Luke Gardner (California Sea Grant)
Grant Amount: $100,000

This project will provide coordination and facilitation support for two groups focused on interagency and industry aquaculture coordination: the California Aquaculture Interagency Working Group and the California Aquaculture Development Committee. This will foster interagency relationships and strengthen the aquaculture management framework for the state of California. Work will also identify information needs, develop standardized templates for reports, and support the development of collaborative documents towards advancing aquaculture in California.

NATIONAL SEA GRANT LAW CENTER (MISSISSIPPI)

Expanding the State Marine Aquaculture Policy Dashboard to include U.S. territories and new policy attributes

Principal Investigator: Stephanie Otts (National Sea Grant Law Center)
Grant Amount: $204,830

This project will conduct legal and policy research to enhance the data in the State Marine Aquaculture Policy Dashboard, which was launched by the National Sea Grant Law Center and Florida State University in 2024 to inform marine aquaculture management. The Dashboard includes data for the 23 marine coastal states for 42 legislative, policy, and management attributes spanning industry regulations, agency authority, site leasing, farming operations, spatial management tools, biosecurity, sustainability, tribal authority, and capacity building. This project will expand the geographic coverage of the Dashboard by adding data for five U.S. territories (American Samoa, Guam, the Northern Mariana Islands, Puerto Rico, and the U.S. Virgin Islands). Additionally, work will expand the Dashboard's policy breadth by adding new data related to the regulation of environmental impacts/benefits and farming operations.

VIRGINIA SEA GRANT

Permitting and regulatory ecosystem for shellfish aquaculture and enhancement programs in federal waters

Principal Investigators: David Rudders (Virginia Institute of Marine Science), Andrew Scheld (Virginia Institute of Marine Science)
Grant Amount: $250,000

The goal of this project is to identify law and policy avenues for facilitating aquaculture-based stock enhancement of Atlantic surf clams and Atlantic sea scallops in federal waters. The project will form an advisory committee composed of individuals representing the fishing, aquaculture, offshore wind industries, state regulators, federal regulators, and academic researchers. This committee will guide a team of legal researchers in the analysis of the legal, policy, and regulatory issues surrounding the potential implementation of aquaculture-based stock enhancement efforts for surf clams and sea scallops in U.S. federal waters.

ENHANCING KNOWLEDGE OF AQUACULTURE TECHNOLOGIES, PRACTICES, AND BUSINESSES

Eight projects based in California, Georgia, Hawai‘i, Illinois, Maine, Massachusetts, and Rhode Island were selected for a total of $557,913 in fiscal year 2025 funding, towards supporting opportunities for researchers, extension personnel, and aquaculture industry members to exchange information. This includes visits and learning at aquaculture-related sites domestically and abroad to enhance knowledge of aquaculture technologies, production practices and systems, businesses, or other associated topics. Learn more about these projects below.

CALIFORNIA SEA GRANT

Knowledge exchange on macroalgae cultivation - Incheon, South Korea, to San Diego, CA, USA

Principal Investigator: Simona Augyte (California Sea Grant)
Grant Amount: $74,253

This project will enhance U.S.seaweed aquaculture on the West Coast by forming a collaboration between researchers, growers, and extension professionals. The collaboration will be led by California Sea Grant and researchers and growers in South Korea, who are leaders in seaweed aquaculture. A site visit to the Marine Ecology and Green Aquaculture Laboratory at Incheon National University in South Korea will facilitate a knowledge exchange and teach attendees cutting-edge seaweed production technologies and methods. These will include seaweed germplasm banking, integrated multi-trophic aquaculture, and strain selection. To share and disseminate the knowledge gained, a workshop and webinar will be held, and a seaweed nursery lab manual and training materials for students will be developed.

CALIFORNIA SEA GRANT

U.S and Japan - Sea urchin roe enhancement knowledge exchange

Principal Investigator: Luke Gardner (California Sea Grant)
Grant Amount: $74,560
This project will offer U.S. researchers, farmers, processors, and divers involved in the development of purple sea urchin aquaculture the opportunity to visit Japanese research facilities and farms involved in sea urchin aquaculture. On these tours, participants will learn advanced techniques used to culture urchins, particularly methods focused on urchin gonad enhancement, as a means of utilizing animals collected from West Coast urchin barrens, which in turn protects kelp forests that wild urchin populations have decimated. Knowledge acquired from the Japanese will be shared with the West Coast and broader U.S. aquaculture communities through outreach efforts, including the development of a white paper and presentations at conferences.

GEORGIA SEA GRANT

Experiential learning of New Zealand aquaculture production by Sea Grant extension agents and commercial shellfish farmers to facilitate aquaculture growth in the U.S.

Principal Investigator: Tom Bliss (Georgia Sea Grant)
Grant Amount: $73,628
This project will allow producers, researchers, and extension personnel associated with the rapidly growing oyster aquaculture industry in the southeastern U.S. to be exposed to and learn about oyster aquaculture as practiced in New Zealand. Participants will undertake site visits to aquaculture facilities in New Zealand, as well as attend the New Zealand Aquaculture Conference. The knowledge acquired by the participants will subsequently be shared with the broader southeastern U.S. oyster aquaculture community through outreach efforts, including the creation of a Story Map documenting the trip and presentations at aquaculture conferences.

HAWAIʻI SEA GRANT

Innovate, cultivate, sustain: Global insights for advancing U.S. aquaculture in Hawaiʻi

Principal Investigators: Cherie Kauahi (Hawaiʻi Sea Grant), Anne Rosa (Hawaiʻi Sea Grant), Bradley ʻKaiʼ Fox (Hawaiʻi Sea Grant), Darren K. Okimoto (Hawaiʻi Sea Grant)

Grant Amount: $75,000

This project will strengthen sustainable aquaculture in Hawaiʻi through an international knowledge exchange with partners in Australia. Focusing on striped mullet, a culturally and economically important finfish species in Hawaiʻi, this project will address challenges in hatchery production, ecosystem-based management, and Indigenous aquaculture practices of this species. Reciprocal site visits with Australian collaborators will be conducted to observe and exchange information regarding successful breeding, monitoring, and stocking strategies. Insights gained will be shared with Hawaiian aquaculture practitioners through outreach efforts, including workshops to inform striped mullet culture practices, particularly those relevant to Hawaiian fishponds.

ILLINOIS-INDIANA SEA GRANT

Enhancing experiential learning and professional growth for largemouth bass producers in the Great Lakes region

Principal Investigators: Stuart Carlton (Illinois-Indiana Sea Grant), Amy Shambach (Illinois-Indiana Sea Grant), Herbert Quintero (Ohio State University)

Grant Amount: $75,000

This project will provide Great Lakes region extension personnel, researchers, and largemouth bass farmers with experiential learning opportunities to improve the production and profitability of their businesses. Participants will visit and interact with vertically integrated largemouth bass aquaculture businesses in California and Washington. Subsequently, workshops focused on largemouth bass breeding, hatching, larval rearing, fingerling rearing, and culture systems will be held to share the knowledge gained from visits. This farmer-to-farmer technology transfer will benefit Great Lakes region largemouth bass producers by providing best practices and practical, cost-effective options, which will help expand the industry.

MAINE SEA GRANT

Advancing sea urchin aquaculture knowledge for industry growth: A Japanese exchange initiative

Principal Investigators: Dana Morse (Maine Sea Grant), Stephen Eddy (University of Maine), Coleen Suckling (University of Rhode Island), Hugh Cowperthwaite (Coastal Enterprises, Inc.)

Grant Amount: $74,746

This project will expose members of the New England aquaculture community to state-of-the-art Japanese urchin farming, harvesting, processing, and marketing facilities, where they will meet and learn from Japanese experts. Participants will include current and prospective sea urchin growers, scientists, hatchery experts, extension professionals, and fishery partners. Knowledge gained from this learning exchange will subsequently be shared with the broader New England and U.S. aquaculture communities through various outreach activities to grow the U.S. sea urchin aquaculture industry.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY SEA GRANT

AquaCulture Shock: International internships for U.S.-Norway aquaculture knowledge exchange

Principal Investigators: Michael Triantafyllou (Massachusetts Institute of Technology Sea Grant), Lily Keyes (MIT Sea Grant), Danny Badger (MIT Sea Grant, badgerd@mit.edu)

Grant Amount: $51,887

This project builds on an international initiative to exchange knowledge on the production of Atlantic salmon through student internships with key partners in Norway, in partnership with MIT International Science and Technology Initiatives. The project will match four MIT Sea Grant students with internships at leading aquaculture-focused labs and companies in Norway, including the Norwegian University of Science and Technology (NTNU) Field Robotics Lab and SINTEF Ocean's aquaculture department. The students will gain knowledge of systems and technologies related to farming salmon, including autonomy for offshore farms, sensors and underwater vehicles for monitoring fish and structure integrity, and machine learning for fish health and feeding. The internships will support the development of an informed aquaculture workforce, further a United States-Norway collaboration in the field of aquaculture, and enhance knowledge of technologies and systems for farming Atlantic salmon in the U.S.

RHODE ISLAND SEA GRANT

Advancing aquaculture in Rhode Island through cross-Atlantic farming knowledge exchange

Principal Investigators: Azure Cygler (Rhode Island Sea Grant), Robbie Hudson (Rhode Island Sea Grant),
Tracey Dalton (Rhode Island Sea Grant)
Grant Amount: $58,839

This project will provide Rhode Island Sea Grant mid-career aquaculture extension professionals the opportunity to enhance their knowledge regarding seaweed and shellfish aquaculture as practiced by global leaders in Europe. Participants will take part in site visits to oyster and seaweed farms in Norway, France, and Ireland to enhance their knowledge on sustainable culture practices, hatchery production, and integrating farm tours into business models to enhance revenue and build community education and support. From information gained, the extension professionals will develop an expanded course curriculum for Rhode Island Sea Grant's Professional Aquaculture Training course, integrating European expertise, which will contribute to workforce development in support of the Rhode Island and adjacent New England aquaculture industries.